The Parallel Multiple Mediator Model with Covariates

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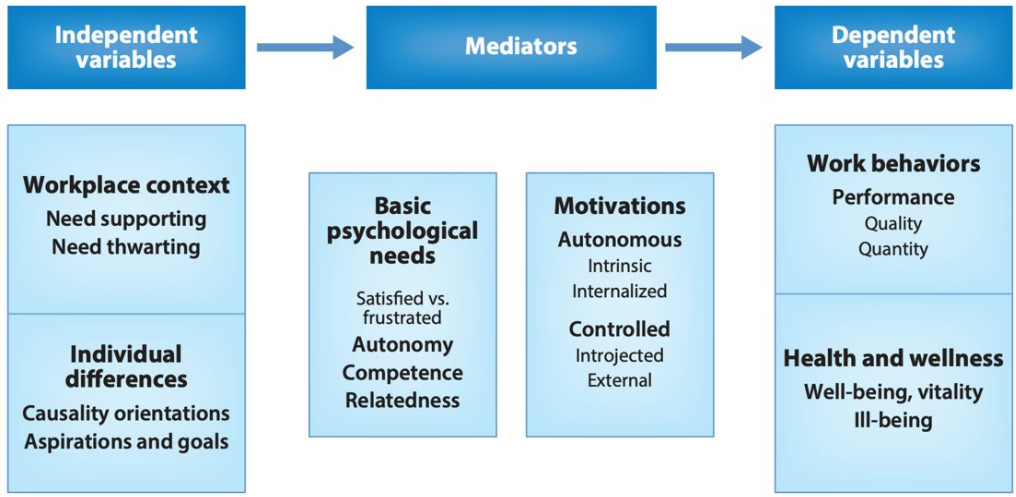
2026-08-01

Problem

Why is it that some people have higher task performance whereas others don't?

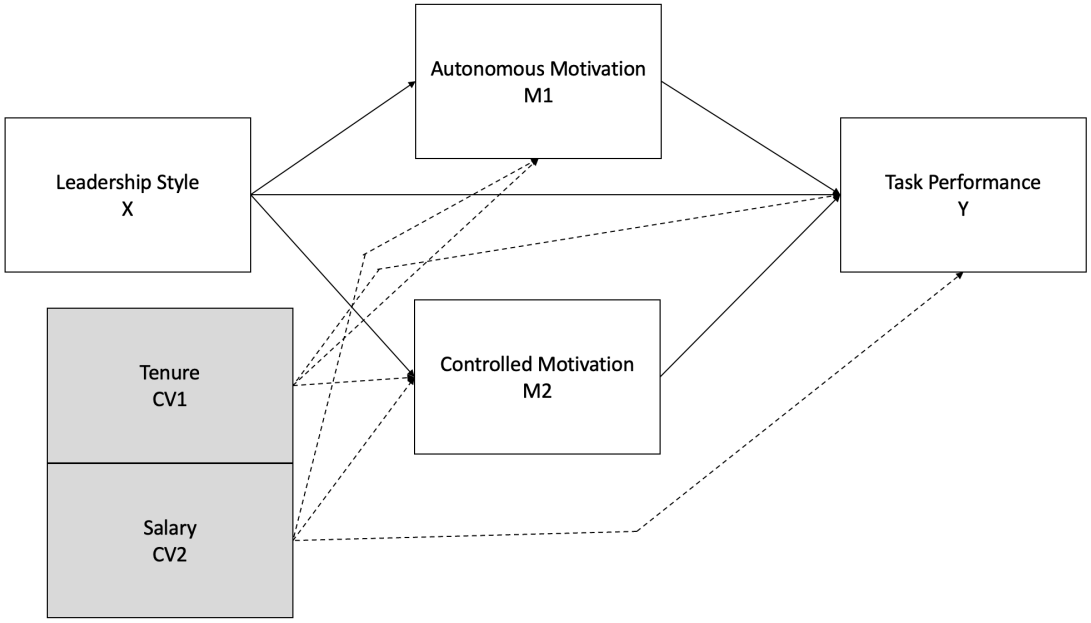
Theory

According to self-determination theory, individuals' work behaviors are predicted by their motivations (autonomous and controlled), which, in turn, are predicted by workplace context (e.g., leadership style). **Furthermore, it is known that tenure and salary are two important antecedents of work motivation and task performance. Hence, we want to control the effects of these two "covariates."**



We want to test if this part of this model is true or not. So, we go out to collect some data but what should we do next?

Notation: Diagram

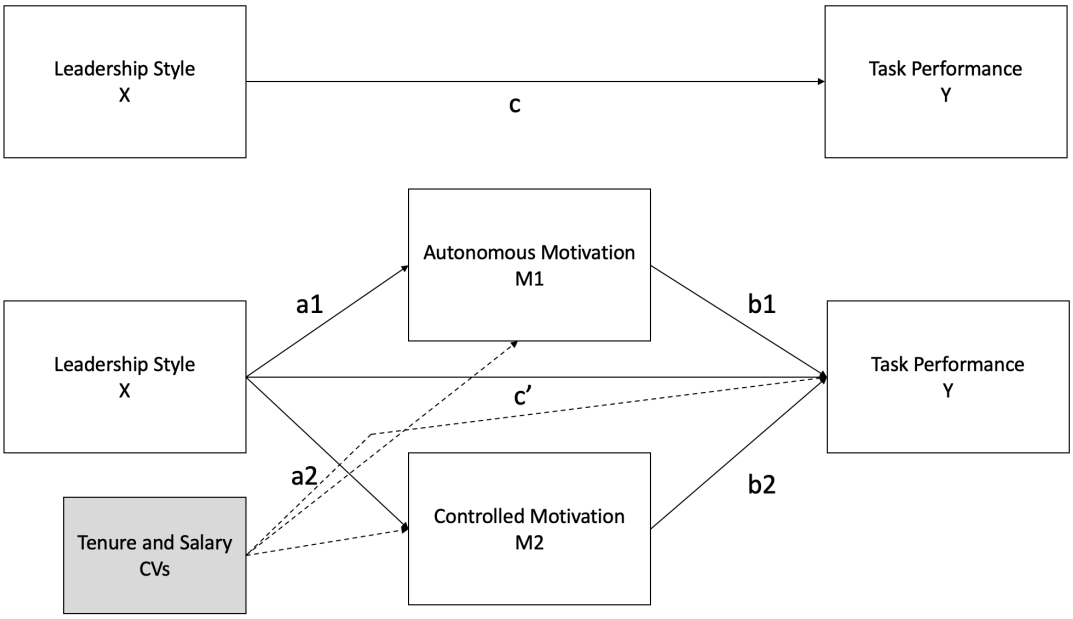


- **X**: causal antecedent variable
- **Y**: consequent variable
- **M1** "mediator 1"
- **M2** "mediator 2"
- **CV1** "covariate 1"
- **CV2** "covariate 2"

Activity

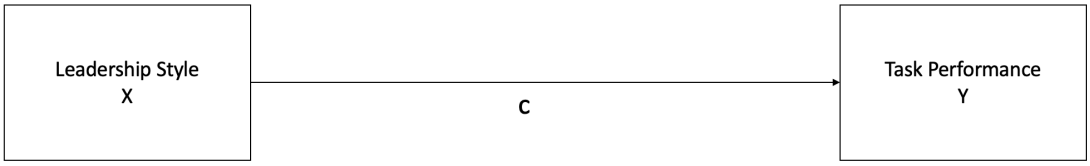
- Use the notation shown here to draw the relation you would like to investigate.

Notation: Total, direct, and indirect effects

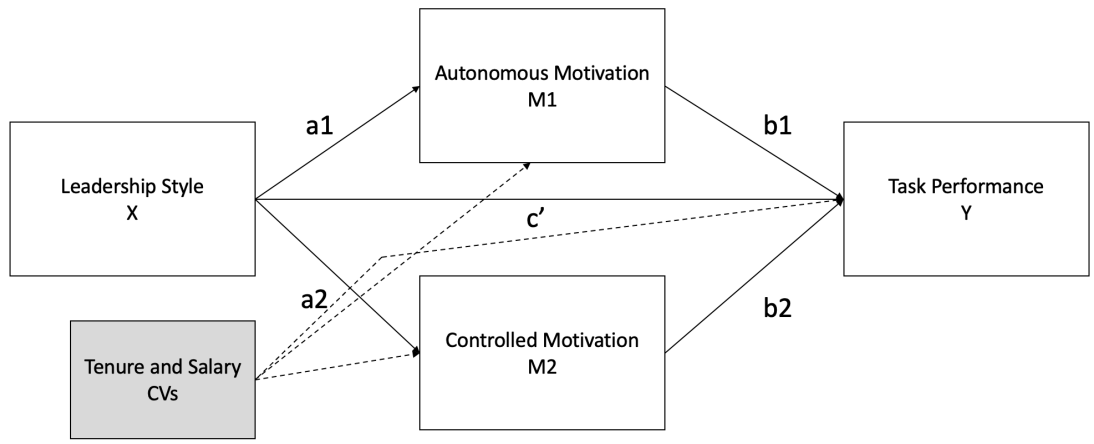


- Let a1, a2, b1, b2, c, and c' be quantifications of causal effects, such as regression coefficients in an ordinary least squares model (or maximum likelihood path estimates in a structural equation model).
- c = "total effect" of X on Y (i.e., the effect of X on Y when we don't account for the effect of M)
- **a1 × b1 = "indirect effect" of X on Y (This is our main focus.)**
- **a2 × b2 = "indirect effect" of X on Y (This is our main focus.)**
- c' = "direct effect" of X on Y
- total effect (c) = direct effect (c') + indirect effect (a1 × b1) + indirect effect (a2 × b2)
- indirect effect (a1 × b1) + indirect effect (a2 × b2) = total effect (c) - direct effect (c')
- We can still do our mediation analysis even if the total effect is zero, **c is not the precursor of the mediation analysis**.

Regression Equation



$$TaskPerformance_j = i + cLeadershipStyle_j + \varepsilon_j$$



$$AutonomousMotivation_j = i + a1LeadershipStyle_j + cv1Tenure_j + cv2Salary_j + \varepsilon_j$$

$$ControlledMotivation_j = i + a2LeadershipStyle_j + cv1Tenure_j + cv2Salary_j + \varepsilon_j$$

$$TaskPerferomance_j = i + b1AutonomousMotivation_j + b2ControlledMotivation_j + c'LeadershipStyle_j + cv1Tenure_j + cv2Salary_j + \varepsilon_j$$

Transformational Leadership

Transformational leadership is a leadership style that can inspire positive changes in those who follow. Transformational leaders are generally energetic, enthusiastic, and passionate. Not only are these leaders concerned and involved in the process, but they are also focused on helping every member of the group succeed.

Run Process Macro

Whenever you would like to run conditional process analysis, don't forget to open and run the script of Process Macro!

```
******************** PROCESS for R Version 4.3.1 ********************

          Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

PROCESS is now ready for use.
Copyright 2020-2023 by Andrew F. Hayes ALL RIGHTS RESERVED
Workshop schedule at http://haskayne.ucalgary.ca/CCRAM
```

Load Packages and Import Data

```
library(tidyverse)
library(psych)
```

```r
library(haven)

data <- read_sav("sdt_cv.sav")

#view our data
data
```

```
# A tibble: 500 × 6
   leader  auto control performance tenure salary
    <dbl> <dbl>   <dbl>       <dbl>  <dbl>  <dbl>
 1   2.15  3.00   2.26         5.29     35 58454.
 2   4.15  4.22   1.70         7.49     39 64509.
 3   2.64  3.16   1.74         4.87     32 57932.
 4   4.53  3.88   1.99         5.91     28 87439.
 5   4.76  4.59   1.84         7.45     39 90299.
 6   1.18  2.07   3.25         3.09     21 84493.
 7   3.11  3.94   2.68         5.60     33 82674.
 8   4.57  4.75   0.849        7.20     41 60100.
 9   3.21  4.81   1.21         6.30     41 55933.
10   2.83  3.38   2.19         5.26     29 64304.
# ℹ 490 more rows
```

```r
#descriptive statistics
describe(data)
```

```
            vars   n     mean       sd   median  trimmed      mad      min
leader         1 500     2.98     1.14     2.91     2.97     1.43     1.00
auto           2 500     3.79     0.91     3.81     3.78     1.02     1.62
control        3 500     2.20     0.89     2.22     2.21     0.92     0.09
performance    4 500     5.41     1.47     5.34     5.37     1.70     1.89
tenure         5 500    29.80     7.36    30.00    29.70     7.41    11.00
salary         6 500 84914.35 21387.52 83809.47 84652.53 27433.44 46238.20
                  max    range  skew kurtosis     se
leader           5.00     4.00  0.10    -1.19   0.05
auto             5.86     4.25  0.03    -0.66   0.04
control          4.40     4.31 -0.07    -0.54   0.04
performance      9.10     7.20  0.17    -0.72   0.07
tenure          49.00    38.00  0.10    -0.56   0.33
salary      129326.68 83088.48  0.07    -1.16 956.48
```

Visualize (Bivariate Correlations)

```r
data |> as.matrix() |> Hmisc::rcorr()
```

```
            leader  auto control performance tenure salary
leader        1.00  0.77   -0.77        0.89   0.56   0.23
auto          0.77  1.00   -0.60        0.81   0.61   0.25
control      -0.77 -0.60    1.00       -0.76  -0.47  -0.16
performance   0.89  0.81   -0.76        1.00   0.65   0.23
tenure        0.56  0.61   -0.47        0.65   1.00   0.21
salary        0.23  0.25   -0.16        0.23   0.21   1.00
```

```
n= 500


P
            leader auto  control performance tenure salary
leader             0e+00 0e+00   0e+00       0e+00  0e+00
auto        0e+00        0e+00   0e+00       0e+00  0e+00
control     0e+00  0e+00         0e+00       0e+00  3e-04
performance 0e+00  0e+00 0e+00               0e+00  0e+00
tenure      0e+00  0e+00 0e+00   0e+00              0e+00
salary      0e+00  0e+00 3e-04   0e+00       0e+00
```

```
#data |> select(leader, auto, control, performance, tenure, salary) |> as.mat
rix() |> Hmisc::rcorr()
```

Model (conditional process analysis: the parallel multiple mediator model)

Unfortunately, there is no existing model we can choose to run the parallel multiple mediator analysis. However, the conditional process analysis framework allows us to create and customize our own model. This is how:

	X	M1	M2
M1	1		
M2	1	0	
Y	1	1	1

```
process(data = data, y = "performance", x = "leader", m = c("auto", "control"
), cov = c("tenure", "salary"), total = 1, seed = 12345, bmatrix = c(1, 1, 0,
 1, 1, 1))


******************** PROCESS for R Version 4.3.1 ********************

           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

Model : CUSTOM
    Y : performance
    X : leader
   M1 : auto
   M2 : control

Covariates:
```

```
       tenure salary

Sample size: 500

Custom seed: 12345


*******************************************************************
Outcome Variable: auto

Model Summary:
          R      R-sq       MSE         F       df1       df2         p
     0.8045    0.6472    0.2963  303.2793    3.0000  496.0000    0.0000

Model:
             coeff        se         t         p      LLCI      ULCI
constant    1.1610    0.1281    9.0650    0.0000    0.9094    1.4127
leader      0.4931    0.0261   18.9157    0.0000    0.4419    0.5444
tenure      0.0324    0.0040    8.0887    0.0000    0.0245    0.0402
salary      0.0000    0.0000    1.9122    0.0564   -0.0000    0.0000

*******************************************************************
Outcome Variable: control

Model Summary:
          R      R-sq       MSE         F       df1       df2         p
     0.7728    0.5972    0.3204  245.1590    3.0000  496.0000    0.0000

Model:
             coeff        se         t         p      LLCI      ULCI
constant    4.0551    0.1332   30.4469    0.0000    3.7934    4.3168
leader     -0.5807    0.0271  -21.4211    0.0000   -0.6340   -0.5275
tenure     -0.0074    0.0042   -1.7885    0.0743   -0.0156    0.0007
salary      0.0000    0.0000    0.9337    0.3509   -0.0000    0.0000

*******************************************************************
Outcome Variable: performance

Model Summary:
          R      R-sq       MSE         F       df1       df2         p
     0.9291    0.8633    0.2999  623.8938    5.0000  494.0000    0.0000

Model:
             coeff        se         t         p      LLCI      ULCI
constant    1.7188    0.2245    7.6564    0.0000    1.2777    2.1599
leader      0.6507    0.0427   15.2541    0.0000    0.5668    0.7345
auto        0.3762    0.0452    8.3291    0.0000    0.2875    0.4650
control    -0.2689    0.0434   -6.1896    0.0000   -0.3542   -0.1835
tenure      0.0310    0.0043    7.2249    0.0000    0.0226    0.0395
```

```
salary     -0.0000    0.0000   -0.1048    0.9166   -0.0000    0.0000

************************ TOTAL EFFECT MODEL **************************
Outcome Variable: performance

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.9129     0.8335     0.3639   827.4266     3.0000   496.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     1.0654     0.1419     7.5064     0.0000     0.7865     1.3442
leader       0.9923     0.0289    34.3498     0.0000     0.9356     1.0491
tenure       0.0452     0.0044    10.1958     0.0000     0.0365     0.0539
salary       0.0000     0.0000     0.3181     0.7505    -0.0000     0.0000

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```
     effect        se         t         p      LLCI      ULCI
     0.9923    0.0289   34.3498    0.0000    0.9356    1.0491

Direct effect of X on Y:
     effect        se         t         p      LLCI      ULCI
     0.6507    0.0427   15.2541    0.0000    0.5668    0.7345

Indirect effect(s) of X on Y:
           Effect    BootSE  BootLLCI  BootULCI
TOTAL      0.3417    0.0368    0.2691    0.4131
auto       0.1855    0.0238    0.1397    0.2338
control    0.1561    0.0268    0.1043    0.2108

******************** ANALYSIS NOTES AND ERRORS ************************

Level of confidence for all confidence intervals in output: 95

Number of bootstraps for percentile bootstrap confidence intervals: 5000

#data = the name of your dataset
#y = consequent variable
#x = causal antecedent variable
#m = mediator
#cov = covariate
#total = whether you want the total effect printed out or not
#seed = for the purpose of replication
#bmatrix = specify how x, m, and y should be related

#evaluation: overall model + predictor(s)

#overall model: Multiple R-squared (ranges from 0-1) helps us evaluate the pr
edictability of our model. The higher the better.

#Outcome Variable: auto => path a1

#interpretation: About 64.72% of variance in autonomous motivation is explain
ed by (transformational) leadership style.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Leadership significantly and positively predicts autonomous
motivation after controlling for the effects of tenure and salary. One unit i
ncrease in leadership is expected to be associated with an increase of 0.4931
 units in autonomous motivation.

#Outcome Variable: control => path a2
```

```
#interpretation: About 59.72% of variance in controlled motivation is explain
ed by (transformational) leadership style.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Leadership significantly and negatively predicts controlled
motivation after controlling for the effects of tenure and salary. One unit i
ncrease in leadership is expected to be associated with a decrease of 0.5807
units in controlled motivation.

#Outcome Variable: performance => path b1, b2, and c'

#interpretation: About 86.33% of variance in task performance is explained by
 leadership, autonomous motivation, AND controlled motivation.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation b1: Autonomous motivation significantly and positively predic
ts task performance after controlling for the effects of tenure and salary. O
ne unit increase in autonomous motivation is expected to be associated with a
n increase of 0.3762 units in task performance.

#interpretation b2: Controlled motivation significantly and negatively predic
ts task performance after controlling for the effects of tenure and salary. O
ne unit increase in controlled motivation is expected to be associated with a
n decrease of 0.2689 units in task performance.

#interpretation c': Leadership significantly and positively predict task perf
ormance after controlling for the effects of tenure and salary. One unit incr
ease in leadership is expected to be associated with an increase of 0.6507 un
its in task performance.

#Outcome Variable: performance => path c

#interpretation: About 83.35% of variance in task performance is explained by
 leadership ALONE.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation c: Leadership significantly and positively predicts task perf
ormance. One unit increase in leadership is expected to be associated with an
 increase of 0.9923 units in task performance.
```

```
#TOTAL, DIRECT, AND INDIRECT EFFECTS OF X ON Y
#Indirect effect(s) of X on Y => a1 x b1 AND a2 x b2
#To evaluate whether M significantly mediates the relation between X and Y, y
ou need to check BootLLCI and BootULCI. If zero is not in the confidence inte
rval, then we can claim an indirect effect different from zero with 95% confi
dence. This is akin to (though not exactly the same as) rejecting the null hy
pothesis of no indirect effect at the alpha = 0.05 level of significance.

#interpretation a1 x b1: Autonomous motivation significantly mediates the rel
ation between leadership and task performance after controlling for the effec
ts of tenure and salary (a1b1 = 0.1855, 95% CI [.1397, .2338]).

#interpretation a1 x b1: Leadership is significantly and indirectly related t
o task performance through the pathway of autonomous motivation (a1b1 = 0.185
5, 95% CI [.1397, .2388]).

#interpretation a2 x b2: Controlled motivation significantly mediates the rel
ation between leadership and task performance after controlling for the effec
ts of tenure and salary (a2b2 = 0.1561, 95% CI [.1043, .2108]).

#interpretation a2 x b2: Leadership is significantly and indirectly related t
o task performance through the pathway of controlled motivation (a2b2 = 0.156
1, 95% CI [.1043, .2108]).

#Conclusion: When employees perceived their supervisors to be more transforma
tional, the higher autonomous motivation they have, which, in turn, is signif
icantly and positively associated with task performance. Also, when employees
 perceived their supervisors to be more transformational, the lower controlle
d motivation they have, which, in turn, is significantly and negatively assoc
iated with task performance.
```

Assignment 5

Use NSCH dataset. Suppose you are interested in studying children teeth condition. Your theory is that age (**SC_AGE_YEARS**) serves as a causal antecedent variable, sugary drink consumption (**SugarDrink_21**), fruit consumption (**fruit_21**), and vegetable consumption (**vegetables_21**) serves as parallel mediators, and the condition of children's teeth (**TeethCond_21**) serves as a consequent variable. Furthermore, the income level of child's household serves as a covariate (**povlev4_21**). Please run a conditional process analysis to find out the answer. (Let's assume that these five variables are measured on the same scale.). **Please submit your R script with interpretations to the designated folder on iStudy. Your R script should be titled as "assignment 5_LastFirstName."**

- Here is the information on those three variables:
  - age (**SC_AGE_YEARS**) : Higher values indicate older children.
  - sugary drink consumption (**SugarDrink_21**) : Higher values indicate that the child have sugary drinks more often on a weekly basis.

- fruit consumption (**fruit_21**) : Higher values indicate that the child eat more fruits on a weekly basis.
- vegetable consumption (**vegetables_21**) : Higher values indicate that the child eat more vegetables on a weekly basis.
- teeth condition (**TeethCond_21**) : Low values indicate better teeth conditions.
- income level of child's household (**povlev4_21**) : Low values indicate better lower income.

- Draw the diagram to represent the paralell multiple mediator model.
- Write down the bmatrix.
- Import the dataset.
- Remove value labels by using the following code (assuming you name the dataset "data"):

```
library(labelled)
val_labels(data) <- NULL
```

- Take a look at your data and descriptive statistics.
- Run Pearson's bivariate correlations
- Run the conditional process analysis
- Find out the effect of path a1, a2, a3, b1, b2, b3, c, c', cvs, a1 x b1, a2 x b2, a3 x b3
- Evaluate the overall model and individual predictor.
- Write down your interpretation of the results for each path (i.e., a1, a2, a3, b1, b2, b3, c, c', a1 x b1, a2 x b2, a3 x b3)
- Is your theory correct? Does sugary drink consumption significantly mediate the relation between age and teeth condition after controlling for the household income level? Does fruit consumption significantly mediate the relation between age and teeth condition after controlling for the household income level? Does vegetable consumption significantly mediate the relation between age and teeth condition after controlling for the household income level? Which path is more strongly associated with teeth condition after controlling for the household income level?

Assignment: Find an empirical journal article

- Find an empirical article in your field that uses the parallel multiple mediator model with covariates.
- What is the hypothesized model? That is, what is the diagram that visualizes the relations?

- Define X, M1, M2, CVs, and Y using layperson's terms
- Find out the values for a1, a2, b1, b2, c', cvs, a1 x b1, and a2 x b2.
- Find out R^2
- Share your interpretation of the results for each path (i.e., a1, a2, b1, b2, c', a1 x b1, a2 x b2) using layperson's terms
- Are the mediation effects significant after controlling for the effects of covariates? How do you know?
- Make a presentation to your classmates with PPT slides.
- Your presentation cannot go over than 5 mins.
- **Submit your PPT slides to the designated folder on iStudy.**